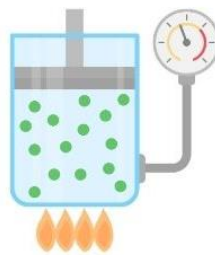


- Q1-** From the concentrations of C_4H_9Cl (butyl chloride) at different times given below, calculate the average rate of the reaction:
 $C_4H_9Cl + H_2O \rightarrow C_4H_9OH + HCl$
 during different intervals of time.
- Q2-** The decomposition of N_2O_5 in CCl_4 at 318K has been studied by monitoring the concentration of N_2O_5 in the solution. Initially the concentration of N_2O_5 is 2.33 mol L^{-1} and after 184 minutes, it is reduced to 2.08 mol L^{-1} . The reaction takes place according to the equation
 $2 N_2O_5 (g) \rightarrow 4 NO_2 (g) + O_2 (g)$
 Calculate the average rate of this reaction in terms of hours, minutes and seconds. What is the rate of production of NO_2 during this period?
- Q3-** For the reaction $R \rightarrow P$, the concentration of a reactant changes from 0.03M to 0.02M in 25 minutes. Calculate the average rate of reaction using units of time both in minutes and seconds. 3.2 In a reaction, $2A \rightarrow \text{Products}$, the concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes. Calculate the rate during this interval?
- Q4-** Calculate the overall order of a reaction which has the rate expression
 (a) $\text{Rate} = k [A]^{1/2} [B]^{3/2}$
 (b) $\text{Rate} = k [A]^{3/2} [B]^{-1}$
- Q5-** Identify the reaction order from each of the following rate constants.
 (i) $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ (ii) $k = 3 \times 10^{-4} \text{ s}^{-1}$
- Q6-** For a reaction, $A + B \rightarrow \text{Product}$; the rate law is given by, $r = k [A]^{1/2} [B]^2$. What is the order of the reaction?
- Q7-** The conversion of molecules X to Y follows second order kinetics. If concentration of X is increased to three times how will it affect the rate of formation of Y ?
- Q8-** The initial concentration of N_2O_5 in the following first order reaction
 $N_2O_5 (g) \rightarrow 2 NO_2 (g) + 1/2 O_2 (g)$ was $1.24 \times 10^{-2} \text{ mol L}^{-1}$ at 318 K. The concentration of N_2O_5 after 60 minutes was $0.20 \times 10^{-2} \text{ mol L}^{-1}$. Calculate the rate constant of the reaction at 318 K.
- Q9-** The following data were obtained during the first order thermal decomposition of $N_2O_5 (g)$ at constant volume:
 $2N_2O_5 (g) \rightarrow 2N_2O_4 (g) + O_2 (g)$
- | S.No. | Time/s | Total Pressure/(atm) |
|-------|--------|----------------------|
| 1. | 0 | 0.5 |
| 2. | 100 | 0.512 |
- Calculate the rate constant
- Q10-** A first order reaction is found to have a rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. Find the half-life of the reaction. Half-life for a first order reaction is

$$t_{1/2} = \frac{0.693}{k}$$

$$t_{1/2} = \frac{0.693}{5.5 \times 10^{-14} \text{ s}^{-1}} = 1.26 \times 10^{13} \text{ s}$$

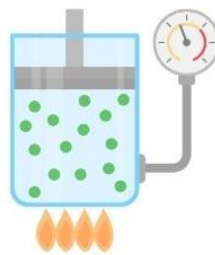
Show that in a first order reaction, time required for completion of 99.9% is 10 times of half-life



($t_{1/2}$) of the reaction.

- Q11-** When reaction is completed 99.9%, $[R]_n = [R]_0 - 0.999[R]_0$
- Q12-** The rate constants of a reaction at 500K and 700K are 0.02s^{-1} and 0.07s^{-1} respectively. Calculate the values of E_a and A .
- Q13-** The first order rate constant for the decomposition of ethyl iodide by the reaction $\text{C}_2\text{H}_5\text{I(g)} \rightarrow \text{C}_2\text{H}_4\text{(g)} + \text{HI(g)}$ at 600K is $1.60 \times 10^{-5}\text{s}^{-1}$. Its energy of activation is 209 kJ/mol. Calculate the rate constant of the reaction at 700K.
- Q14-** What will be the effect of temperature on rate constant ?
- Q15-** The rate of the chemical reaction doubles for an increase of 10K in absolute temperature from 298K. Calculate E_a .
- Q16-** The activation energy for the reaction $2\text{HI(g)} \rightarrow \text{H}_2 + \text{I}_2\text{(g)}$ is 209.5kJ mol^{-1} at 581K. Calculate the fraction of molecules of reactants having energy equal to or greater than activation energy?
- Q17-** From the rate expression for the following reactions, determine their order of reaction and the dimensions of the rate constants.
- (i) $3\text{NO(g)} \rightarrow \text{N}_2\text{O(g)}$ Rate = $k[\text{NO}]^2$
- (ii) $\text{H}_2\text{O}_2\text{(aq)} + 3\text{I}^-\text{(aq)} + 2\text{H}^+ \rightarrow 2\text{H}_2\text{O(l)} + \text{I}_3^-$ Rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$
- (iii) $\text{CH}_3\text{CHO(g)} \rightarrow \text{CH}_4\text{(g)} + \text{CO(g)}$ Rate = $k[\text{CH}_3\text{CHO}]^{3/2}$
- (iv) $\text{C}_2\text{H}_5\text{Cl(g)} \rightarrow \text{C}_2\text{H}_4\text{(g)} + \text{HCl(g)}$ Rate = $k[\text{C}_2\text{H}_5\text{Cl}]$
- Q18-** For the reaction: $2\text{A} + \text{B} \rightarrow \text{A}_2\text{B}$ the rate = $k[\text{A}][\text{B}]^2$ with $k = 2.0 \times 10^{-6}\text{mol}^{-2}\text{L}^2\text{s}^{-1}$. Calculate the initial rate of the reaction when $[\text{A}] = 0.1\text{mol L}^{-1}$, $[\text{B}] = 0.2\text{mol L}^{-1}$. Calculate the rate of reaction after $[\text{A}]$ is reduced to 0.06mol L^{-1} .
- Q19-** The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if $k = 2.5 \times 10^{-4}\text{mol}^{-1}\text{L s}^{-1}$?
- Q20-** The decomposition of dimethyl ether leads to the formation of CH_4 , H_2 and CO and the reaction rate is given by
Rate = $k[\text{CH}_3\text{OCH}_3]^{3/2}$
The rate of reaction is followed by increase in pressure in a closed vessel, so the rate can also be expressed in terms of the partial pressure of dimethyl ether, i.e., $(P)^{3/2}$
Rate = $k(P_{\text{CH}_3\text{OCH}_3})^{3/2}$ If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants?
- Q21-** Mention the factors that affect the rate of a chemical reaction.
- Q22-** A reaction is second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is (i) doubled (ii) reduced to half ?
- Q23-** What is the effect of temperature on the rate constant of a reaction? How can this effect of temperature on rate constant be represented quantitatively?
- Q24-** In a pseudo first order reaction in water, the following results were obtained:

t/s	0	30	60	90
[A]/ mol L ⁻¹	0.55	0.31	0.17	0.085



Calculate the average rate of reaction between the time interval 30 to 60 seconds.

- Q25-** A reaction is first order in A and second order in B.
 (i) Write the differential rate equation.
 (ii) How is the rate affected on increasing the concentration of B three times?
 (iii) How is the rate affected when the concentrations of both A and B are doubled?
- Q26-** In a reaction between A and B, the initial rate of reaction (r_0) was measured for different initial concentrations of A and B as given below:

A/ mol L ⁻¹	0.20	0.20	0.40
B/ mol L ⁻¹	0.30	0.10	0.05
r_0 /mol L ⁻¹ s ⁻¹	5.07×10^{-5}	5.07×10^{-5}	1.43×10^{-4}

What is the order of the reaction with respect to A and B?

- Q27-** The following results have been obtained during the kinetic studies of the reaction:
 $2A + B \rightarrow C + D$

Experiment	[A]/mol L ⁻¹	[B]/mol L ⁻¹	Initial rate of formation of D/mol L ⁻¹ min ⁻¹
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

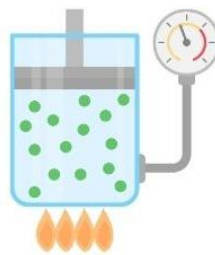
Calculate the average rate of reaction between the time interval 30 to 60 seconds.

- Q28-** The reaction between A and B is first order with respect to A and zero order with respect to B. Fill in the blanks in the following table:

Experiment	[A]/ mol L ⁻¹	[B]/ mol L ⁻¹	Initial rate/mol L ⁻¹ min ⁻¹
I	0.1	0.1	2.0×10^{-2}
II	–	0.2	4.0×10^{-2}
III	0.4	0.4	–
IV	–	0.2	2.0×10^{-2}

- Q29-** Calculate the half-life of a first order reaction from their rate constants given below: (i) 200 s⁻¹ (ii) 2 min⁻¹ (iii) 4 years⁻¹
- Q30-** The half-life for radioactive decay of ¹⁴C is 5730 years. An archaeological artifact containing wood had only 80% of the ¹⁴C found in a living tree. Estimate the age of the sample.
- Q31-** The experimental data for decomposition of N₂O₅
 $[2N_2O_5 \rightarrow 4NO_2 + O_2]$
 in gas phase at 318K are given below:

t/s	0	400	800	1200	1600	2000	2400	2800	3200
$10^2 \times [N_2O_5]/$ mol L ⁻¹	1.63	1.36	1.14	0.93	0.78	0.64	0.53	0.43	0.35



- (i) Plot $[N_2O_5]$ against t .
- (ii) Find the half-life period for the reaction.
- (iii) Draw a graph between $\log[N_2O_5]$ and t .
- (iv) What is the rate law ?
- (v) Calculate the rate constant.
- (vi) Calculate the half-life period from k and compare it with (ii).

- Q32-** The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $1/16$ th value?
- Q33-** During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If 1mg of ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically.
- Q34-** For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction.
- Q35-** A first order reaction takes 40 min for 30% decomposition. Calculate $t_{1/2}$.
- Q36-** For the decomposition of azoisopropane to hexane and nitrogen at 543 K, the following data are obtained.

t (sec)	$P(\text{mm of Hg})$
0	35.0
360	54.0
720	63.0

Calculate the rate constant.

- Q37-** The following data were obtained during the first order thermal decomposition of SO_2Cl_2 at a constant volume. $\text{SO}_2\text{Cl}_2(\text{g}) \rightarrow \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$

Experiment	Time/ s^{-1}	Total pressure/atm
1	0	0.5
2	100	0.6

Calculate the rate of the reaction when total pressure is 0.65 atm.

- Q38-** The rate constant for the decomposition of N_2O_5 at various temperatures is given below:

$T/^\circ\text{C}$	0	20	40	60	80
$10^5 \times k/\text{s}^{-1}$	0.0787	1.70	25.7	178	2140

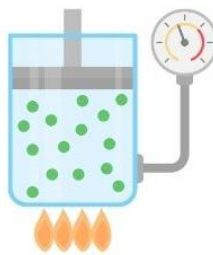
Draw a graph between $\ln k$ and $1/T$ and calculate the values of A and E_a . Predict the rate constant at 30°C and 50°C .

- Q39-** The rate constant for the decomposition of hydrocarbons is $2.418 \times 10^{-5} \text{ s}^{-1}$ at 546 K. If the energy of activation is 179.9 kJ/mol, what will be the value of pre-exponential factor.
- Q40-** Consider a certain reaction $A \rightarrow \text{Products}$ with $k = 2.0 \times 10^{-2} \text{ s}^{-1}$. Calculate the concentration of A remaining after 100 s if the initial concentration of A is 1.0 mol L^{-1} .



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Chapter:- CHEMICAL KINETICS



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- Q41-** Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law, with $t_{1/2} = 3.00$ hours. What fraction of sample of sucrose remains after 8 hours ?
- Q42-** The decomposition of hydrocarbon follows the equation $k = (4.5 \times 10^{11} \text{s}^{-1}) e^{-28000\text{K}/T}$ Calculate E_a .
- Q43-** The rate constant for the first order decomposition of H_2O_2 is given by the following equation: $\log k = 14.34 - 1.25 \times 10^4 \text{K}/T$ Calculate E_a for this reaction and at what temperature will its half-period be 256 minutes?
- Q44-** The decomposition of A into product has value of k as $4.5 \times 10^3 \text{s}^{-1}$ at 10°C and energy of activation 60 kJ mol^{-1} . At what temperature would k be $1.5 \times 10^4 \text{s}^{-1}$?
- Q45-** The time required for 10% completion of a first order reaction at 298K is equal to that required for its 25% completion at 308K. If the value of A is $4 \times 10^{10} \text{s}^{-1}$. Calculate k at 318K and E_a .
- Q46-** The rate of a reaction quadruples when the temperature changes from 293 K to 313 K. Calculate the energy of activation of the reaction assuming that it does not change with temperature.

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